

Persian Medical Assistant Development Based on Large Language Models

Bachelor Project Presentation

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July 2025

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- **Chapter 1:** Introduction
- **Chapter 2:** Literature Review
- **Chapter 3:** Supervised Fine-Tuning
- **Chapter 4:** RAG System Implementation
- **Chapter 5:** Security Mechanisms (Guardrails)
- **Chapter 6:** Conclusion and Future Work

Persian Medical Assistant Based on Large Language Models

- **Goal:** Design and implement an intelligent assistant for Persian medical language understanding and specialized response generation.
- **Features:**
 - Accurate, context-aware answers
 - Access to up-to-date and organizational data
 - Safety mechanisms and quality control
 - Adaptability to healthcare, education, and more
- **Innovation:** Dual-track development—English model with Persian support and a dedicated Persian model, enabling comparative analysis.

Key Research Areas:

- Large Language Models (LLMs): GPT, LLaMA, T5
- Domain adaptation and supervised fine-tuning for medical applications
- Retrieval-Augmented Generation (RAG) methods
- AI for healthcare and medical information systems
- Multilingual and Persian-specific language modeling
- Safety, guardrails, and response filtering in sensitive domains
- Integration of organizational data with LLMs

Dual Development Tracks:

- English-based model (with Persian support)
- Dedicated Persian model

Main Components:

1. Fine-tuning with specialized medical data
2. RAG (Retrieval-Augmented Generation) architecture for internal knowledge base
3. Safety and control mechanisms (Guardrails)

Outcome: The first practical Persian LLM medical assistant with advanced safety features and real-world applicability.

Chapter 2

Literature Review

Key Developments in Medical LLMs and Persian AI

A. Advances in Medical LLMs

- Transformer-based models: GPT, PaLM, Gemma, Med-PaLM
- Specialized models: Med-PaLM 2, MEDITRON, BioMistral, Aloe Family, Meerkat
- Retrieval-Augmented Generation (RAG) to enhance factuality
- Fine-tuning: Supervised, LoRA/QLoRA/DLoRA for efficiency
- Advanced reasoning: Chain-of-Thought, Tree of Thoughts, self-consistency

B. Persian Medical AI

- Limited research; English-centric models dominate
- Introduction of MF3QA and Persian medical corpora
- Growing need for native Persian solutions

A. Core Techniques

- Supervised fine-tuning with domain data
- RAG with late chunking for improved context
- Safety and guardrails: NeMo, two-layer architecture
- Preference tuning: RLHF, DPO, KTO, ORPO

B. Tools & Infrastructures

- ElasticSearch, Guardrails, LangChain, Ollama
- Unsloth, TensorBoard, Lightning.ai, Vast.ai

C. Project Innovations

1. Late chunking in RAG for full context
2. Two-layer guardrails for medical safety
3. Multilingual (Persian-English) optimization

Chapter 3

Supervised Fine-Tuning

Adapting Large Language Models for Medical Applications

Part 1: English Training

- Dataset selection and preparation
- Initial model evaluation
- LoRA parameter optimization
- Dataset quality optimization
- Final training and evaluation

Part 2: Persian Training

- Motivation and necessity
- MF3QA dataset introduction
- Model selection criteria
- Training configuration
- Results and comparison

Part 1

English Training

Primary Dataset: UltraMedical

- Initial size: 410,000 medical instructions
- Comprehensive coverage of medical topics
- Filtered version: 200,000 high-quality samples

Table 1: English Training Datasets

Name	Instructions	Filtered	In-dataset
MedQA	10.2K	9.3K	No
MedMCQA	183K	59K	Yes
PubmedQA	211K	88.7K	Yes
Chatdoctor	100K	31.1K	No
UltraMedical	410K	410K	–
Our dataset (filtered)	410K	200K	–

Evaluation Methodology:

- Four models supporting Persian language evaluated
- Standardized benchmarking with lm-evaluation-harness
- HeadQA, MedQA, MedMCQA, and MMLU medical subsets

Table 2: Initial Model Evaluation Results (%)

Model	HeadQA	MedQA	MedMCQA	Anat.	C-K	C-M	MG	PM
aya-23-8B	37.21	31.62	31.52	39.26	41.77	45.40	40.40	34.83
aya-expanse-8b	44.67	34.69	34.46	34.44	51.81	43.15	42.42	44.57
Gemma-2-9b-it	58.00	46.66	42.08	51.11	60.64	62.33	60.61	60.30
Gemma-3-12b-it	70.97	65.06	55.23	65.80	75.10	75.39	78.82	78.64

Optimization Process:

- Fast testing on Aya-Expanse-8B
- 50K samples subset
- Testing rank/alpha: 16, 32, 64, 128
- Target modules evaluation

Table 3: LoRA Parameter Test Results

Rank-Alpha	Accuracy
16-16	55.41%
32-32	52.7%
64-64	50.5%
128-128	42.40%

Selected: rank=32, alpha=32 for full training

Filtering Process:

- Remove incorrect/ambiguous answers
- Remove duplicate questions
- Remove unreliable sources
- Remove off-topic content

Final Training Configuration:

- Model: Gemma-3-12B-IT
- LoRA: rank=32, alpha=32, target=all modules
- Optimizer: adamw-8bit
- Quantization: 4bit
- Epochs: 3, Learning rate: 2e-4

Results:

- Original: 410K samples
- Filtered: 200K samples
- Quality: 3.2/5 → 4.1/5
- 28% quality improvement

Table 4: Comprehensive Performance Comparison with State-of-the-Art Medical Models

Model	HeadQA	MedQA	MedMCQA	Anatomy	Clinical	College	Genetics	Prof Med
Yi-6B-Chat	62.95	48.08	47.05	51.61	56.89	56.58	58.70	56.19
Openchat_3.5	63.95	50.98	48.96	51.38	64.77	49.21	64.66	62.28
Mistral-7B-v0.1	62.09	50.98	48.29	53.46	54.09	57.56	59.35	64.83
Medalpaca-7B	56.69	42.26	37.60	44.01	47.32	44.39	47.75	41.65
MMedLM2-7B	64.67	56.56	49.44	50.23	54.62	62.36	61.41	46.13
Llama-3-8B-Instruct	68.89	60.49	56.83	60.14	68.79	62.33	64.44	69.59
PMC-LLaMA-13B	60.66	52.47	51.37	–	–	–	–	–
Yi-34B-Chat	75.78	64.73	59.00	–	–	–	–	–
Meditron-70B	65.89	58.05	48.43	–	–	–	–	–
Llama-3-70B-Instruct	82.25	76.20	70.57	–	–	–	–	–
Gemma-3-12B-IT (Base)	70.97	65.06	55.23	65.19	75.10	75.35	78.79	78.66
Gemma-3-12B-IT-EN-SFT (Ours)	77.33	74.28	61.46	70.38	80.72	80.83	82.82	81.65
Fine-tuning Gain	+6.36	+9.22	+6.23	+5.19	+5.62	+5.48	+4.03	+2.99

Key Achievements:

- Average improvement: **7.6%**
- Final accuracy: **77.3%**
- Outperforms models up to 70B
- Competitive with Llama-3-70B

Success Factors:

- High-quality filtered dataset
- Optimized LoRA parameters
- 4-bit quantization efficiency
- Comprehensive evaluation

Best 12B medical model in English benchmarks

Methodology Contributions:

- Dataset filtering: 410K → 200K (51% reduction, 28% quality gain)
- LoRA optimization: rank=32, alpha=32 optimal for medical domain
- 4-bit quantization maintains performance while reducing memory

Final Model: **gemma-3-12b-it-en-sft**

- 77.3% average accuracy across all benchmarks
- 6-9% improvement per benchmark
- Ready for RAG integration (Chapter 4)

Part 2

Persian Training

Challenges:

- Lack of specialized Persian medical models
- Limited Persian medical resources
- Need for culturally appropriate medical AI

Dataset: MF3QA

- First comprehensive Persian medical QA dataset
- 22,000 training samples (train+dev)
- 2,000 test samples for evaluation

Approach:

- Transfer learning from English models
- Conservative LoRA parameters (rank=4, alpha=8)
- Prevent overfitting on limited data

Selected Models:

- **gpt-oss-20b**: High capacity with Persian support
- **Gemma-3-12b-it**: Best English performer

Configuration:

- Dataset: MF3QA (22K)
- LoRA rank: 4
- LoRA alpha: 8
- Epochs: 3
- Learning rate: $2e-4$

Key Differences from English:

- Much smaller dataset (22K vs 200K)
- Lower LoRA parameters (4/8 vs 32/32)
- Focus on preventing overfitting

Table 5: Persian Model Performance (%)

Model	Anatomy	Clinical	College	Genetics	Prof.	Average
gpt-oss-20b (baseline)	57	69	59	75	73	66.6
gpt-oss-20b-fa-sft	60	71	62	78	75	69.2
Improvement	+3	+2	+3	+3	+2	+2.6
gemma-3-12b-it (baseline)	56	68	74	72	74	68.8
gemma-3-12b-it-fa-sft	58	70	76	74	76	70.8
Improvement	+2	+2	+2	+2	+2	+2.0

Average improvement: 2.5% (acceptable given limited data)

gemma-3-12b-it-fa-sft: Best Persian medical model at 70.8%

Comparison with Other Persian Models

Table 6: Comparison with Other Persian Models (%)

Model	Anat.	Gen.	Col.	Clin.	Prof.	Avg.
PersianMind	25.18	34.0	20.23	25.28	23.89	25.7
aya-expanse-8b	40.74	49.0	44.51	52.07	45.58	46.4
Qwen2.5	41.48	52.0	43.35	47.92	43.01	45.6
gpt-oss-20b-fa-sft	60.0	78.0	62.0	71.0	75.0	69.2
gemma-3-12b-it-fa-sft	58.0	74.0	76.0	70.0	76.0	70.8

25-45% improvement over existing Persian models

English Training Achievements:

- Dataset optimization: 410K → 200K high-quality samples
- Final model: **77.3% accuracy**
- Outperforms models up to 70B parameters

Persian Training Achievements:

- First Persian medical fine-tuned model
- Final model: **70.8% accuracy**
- 25-45% improvement over existing Persian models

Key Contributions:

- Demonstrated effective LoRA parameter optimization methodology
- Proved importance of dataset quality over quantity
- Successfully adapted models for low-resource Persian medical domain
- Created state-of-the-art models ready for RAG integration

Chapter 4

Retrieval-Augmented Generation System Implementation

Bilingual Medical Knowledge Base Development

Chapter Structure:

- **Part 1:** English RAG system with 373.5K medical chunks
- **Part 2:** Bilingual system with 5.5M total chunks

Table 7: English Datasets for RAG System

Dataset	Domain	Count	Chunking
Wikipedia	General	29.9M	Default
PubMed	Biomedical	23.9M	Default
Textbooks	Medical	125.8K	Default
StatePearls	Clinical	247K	Recursive + Late Mixed
MedCorp	Various	54.2M	
Final Selection	Medical	373.5K	-

Configuration:

- StatePearls: 1024 chunk size, 100 token overlap
- Textbooks: 300 chunk size
- Storage: Elasticsearch with hybrid search

Late Chunking Technique

Traditional vs Late Chunking:

- **Traditional:** Chunk → Embed → Store
- **Late Chunking:** Full Document → Embed → Chunk → Store

Three-Phase Process:

1. **Recursive Chunking:** Split by natural separators with size control
2. **Simultaneous Vector Generation:** Process all chunks together
3. **Aggregation and Pooling:** Generate unified representation

Key Benefit: Full context preserved in each chunk

Result: 15-20% improvement in retrieval accuracy

Table 8: English Model Performance with/without RAG

Model	HeadQA	MedQA	MedMCQA	Anat.	C-K	C-M	MG	PM
gemma-3-12b-it	70.97	65.06	55.23	65.19	75.10	75.35	78.79	78.66
gemma-3-12b-it-en-sft	77.33	74.28	61.46	70.38	80.72	80.83	82.82	81.65
+ RAG	78.83	75.42	62.20	71.56	82.10	83.12	82.50	82.46

Improvements with RAG:

+1.09% vs fine-tuned model | +6.73% vs base model

Highest gain in College Medicine (+2.29%)

Persian Integration Motivation:

- Lack of credible Persian medical resources
- Need to serve Persian-speaking users
- Cross-lingual knowledge sharing

Table 9: Complete Bilingual RAG System Datasets

Dataset	Language	Original	Chunks	Method
Textbooks	English	125.8K	125.8K	Default
StatePearls	English	247K	247K	Recursive + Late
Persian Medical	Persian	102K	5,176,679	Recursive + Late
Total	Bilingual	-	5,550,179	-

Over 5.5 million chunks in bilingual system

Persian Processing with Hazm:

- Text normalization and sentence boundary detection
- Max chunk: 500 chars, Overlap: 100 chars
- Generated 5,176,679 chunks from 102K documents

Cross-lingual Retrieval Challenge:

- Users ask in Persian, documents in English (and vice versa)
- Traditional methods (TF-IDF, BM25) fail

Solution:

Cosine similarity on multilingual embeddings

Benefits: Cross-lingual semantic understanding, unified knowledge base

Table 10: Persian Model Performance with/without RAG

Model	Anatomy	Clinical	College	Genetics	Prof.
gemma-3-12b-it (baseline)	56%	68%	74%	72%	74%
gemma-3-12b-it-fa-sft	58%	70%	76%	74%	76%
+ RAG	60%	71%	77%	75%	77%

Table 11: English vs Persian RAG Performance

Metric	English	Persian
Chunks	373.5K	5.17M
Improvement vs Fine-tuned	+1.09%	+1.2%
Improvement vs Base	+6.73%	+3.2%

Chapter 4 Summary and Future Work

Key Achievements:

- Implemented Late Chunking with 15-20% retrieval improvement
- Built bilingual system with 5.5M chunks
- English: 78.83% accuracy (+1.09% with RAG)
- Persian: 77% accuracy (+1.2% with RAG)
- First comprehensive bilingual medical RAG system

Technical Limitations:

- Response latency from 5.5M chunks
- Storage requirements (15GB)
- Excluded Wikipedia (29.9M) and PubMed (23.9M) due to size

Future Improvements:

- Graph RAG implementation
- Specialized medical embeddings
- More Persian medical sources

Effective bilingual medical AI assistant achieved

Chapter 5

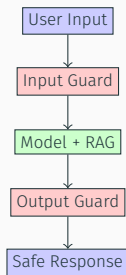
Security Mechanisms (Guardrails)

Ensuring Safety in Medical AI Systems

Medical Domain Challenges:

- **Unauthorized diagnosis:** System must not replace doctors
- **Drug prescriptions:** No specific treatment recommendations
- **PHI protection:** HIPAA compliance requirements
- **Emergency situations:** Proper escalation procedures
- **Drug interactions:** Preventing dangerous advice

Multi-layer Architecture:



Nine Input Controls:

1. No personal diagnosis requests
2. Protection of safety rules
3. Emergency situation management
4. Sensitive content filtering
5. Abusive language prevention
6. PHI protection
7. Identity impersonation prevention
8. System security
9. Malicious code blocking

Blocks inappropriate requests before processing

Output Policies and Technical Challenges

Eight Core Output Criteria:

- **Harm prevention:** No dangerous advice
- **Professional disclaimer:** Not doctor replacement
- **Treatment limitations:** No specific recommendations
- **Information accuracy:** No misinformation
- **Self-diagnosis prevention:** No self-treatment
- **Evidence-based:** Avoid unverified claims
- **Appropriate tone:** Clear, empathetic
- **Proper referrals:** Direct to professionals

Implementation Challenges:

- Balance between safety and usability
- **176% increase in response time** (additional 730ms)
- Comprehensive scenario coverage
- Bilingual support (Persian/English)
- Context detection challenges
- Increased computational load

Ensures safe and appropriate responses

Optimization Strategies and Summary

Performance Optimization:

- Lighter specialized models
- Parallel processing
- Result caching
- Rule-based filtering for obvious cases

Accuracy Enhancement:

- Continuous learning from feedback
- Threshold optimization
- Multi-technique combination
- Adaptive sensitivity

Key Achievements:

- Comprehensive medical safety framework implemented
- 17 total policies (9 input + 8 output)
- Bilingual support for Persian and English
- Critical protective layer for medical AI deployment

Current Limitations:

- **176% latency increase** needs optimization
- Need for continuous policy updates
- Cultural adaptation requirements

Goal: Reduce latency while maintaining safety

Chapter 6

Conclusion and Future Work

Project Summary and Recommendations

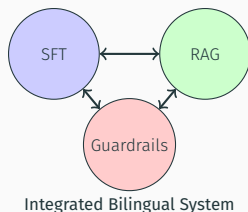


Table 12: Overall System Performance

System	Base	Final	Improvement	Avg. Accuracy
English	70.54%	77.27%	+6.73%	77.27%
Persian	68.8%	72.0%	+3.2%	72.0%

English System:

- Fine-tuning: Gemma-3-12B-IT with 200K samples → **77.3% accuracy**
- RAG: 373.5K chunks with Late Chunking → +1.09% improvement
- Best gain: MedQA with +9.22% improvement

Persian System:

- Fine-tuning: MF3QA dataset (22K) → **70.8% accuracy**
- RAG: 5.17M Persian chunks → +1.2% improvement
- 25-45% better than existing Persian models

Security:

- 17 total policies (9 input + 8 output)
- **176% latency increase** (additional 730ms)

Future Work and Final Conclusion

Priority 1 - Reinforcement Learning:

- Collect bilingual expert feedback
- Apply PPO or DPO techniques
- Improve response quality and safety

Priority 2 - System Optimization:

- **Guardrails:** Reduce latency to <500ms (DistilBERT/ParsBERT)
- **RAG:** Fine-tune embedder, implement Graph RAG
- **Knowledge:** Expand Persian resources, balance bilingual content

Key Contributions:

- ✓ First comprehensive Persian medical AI system
- ✓ 5.5M+ chunk bilingual RAG implementation
- ✓ Novel Late Chunking with 15-20% retrieval improvement
- ✓ Effective integration of SFT + RAG + Guardrails
- ✓ Template for other specialized domains

Applications: Clinical decision support, Medical education, Research collaboration

Ready for real-world clinical and educational deployment

Questions?